

Relative survival of patients with uveal melanoma managed in a single center

Josep Maria Caminal^a, Josepa Ribes^{b,c}, Ramon Clèries^{b,c}, Nuria Ibáñez^a, Luis Arias^a, Josep Maria Piulats^d, Joan Pera^e, Cristina Gutierrez^e and Jorge Arruga^a

To assess the 5-year relative survival of patients diagnosed with uveal melanoma (UM) in a single center. UM patients were recruited from 1995 to 2004 ($N=155$) and were followed until December 2008. Relative survival (RS) methods were used to assess excess mortality. An RS regression model was fitted by sex, age, tumor origin, treatment, and tumor size to estimate the excess hazard rate (EHR) of death from UM. The overall 5-year RS was 90%, lower in women (84.6%) than in men (100%), lower in patients older than 60 years (88.8%) compared with those younger than or of 60 years of age (94.8%). Large tumors (80.8%) showed lower RS than medium (95.1%) and small ones (98.3%). Enucleated patients (80.5%) had lower RS compared with those who received brachytherapy (93.6%) and other treatments (94.7%). A significant EHR was found for women (EHR: 3.65), patients older than 60 years (EHR: 2.25), large-sized melanoma (EHR: 2.45), and during the third (EHR: 5.37) and fourth year (EHR: 3.01) of follow-up. This is the first Spanish study in a single center reporting RS among UM patients, taking into

account clinical characteristics. Prognostic factors that explained RS among UM patients were sex, age, tumor size, and the year of follow-up. We also found a peak of excess mortality from the third until the fourth year after diagnosis, which warrants strict follow-up of these patients during this time interval. *Melanoma Res* 22:271–277 © 2012 Wolters Kluwer Health | Lippincott Williams & Wilkins.

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^aDepartment of Ophthalmology, Bellvitge University Hospital, ^bCatalan Cancer Registry, Oncology Plan for Catalonia, IDIBELL, ^cDepartment of Clinical Sciences, University of Barcelona, Departments of ^dOncology and ^eRadiation Oncology, Catalan Institute of Oncology, Barcelona, Spain

Correspondence to Josepa Ribes MD, PhD, Catalan Cancer Registry, IDIBELL Gran via s/n Km 2.7, L'Hospitalet de Llobregat, Barcelona 08907, Spain
Tel: +34 932 607 124; fax: +34 932 607 981;
e-mail: j.ribes@iconcologia.net

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Introduction

Uveal melanoma (UM) accounts for ~80% of primary ocular malignancies in adults. Recent trends of UM incidence in cancer registries have not shown any marked change in rate over time [1].

It is a fatal disease with a 5- and 10-year melanoma-specific survival of 69–86% and 52–78%, respectively, according to different publications [2–4].

The primary reason for treating UMs is to prevent the metastatic spread of the tumor from the eye to other parts of the body. About 50% of all patients develop metastatic disease, with death in the long-term follow-up after diagnosis, and several case reports have described unusual UMs that have shown clinical metastasis more than 40 years after enucleation [5–8]. In the past, enucleation was the only treatment for UM. Several studies have not found significant differences in survival between enucleated patients and those treated by conservative treatments such as brachytherapy [4,9]. At present, conservative treatments are the preferred therapies in the majority of referral centers because of their safety and predictable results.

When we study the UM survival, often, cause-specific mortality is used, but the risk of misclassification of the underlying cause of death is a major problem. An increased risk of second malignancies [10,11] and misclassified UM metastases as cutaneous melanoma or primary liver cancer [12] have been postulated in UM patients. Therefore, instead of calculating the probability of surviving a specific cancer considering deaths from other causes as censoring events, the concept of relative survival (RS) was developed by comparing the observed survival probability of a group of cancer patients with the survival of a 'similar' cancer-free group. In other words, RS represents the survival experience of a group of patients affected by a malignant tumor adjusted for their probability of dying from all other causes [13–16]. The RS survival can be regarded as the disease-specific survival if the survival in the general population can be assumed to reflect survival from causes other than the disease, and death from the disease is independent of death from other causes.

The aim of our study was to assess the 5-year RS of UM among 155 consecutive patients diagnosed and treated

in Bellvitge University Hospital during the period 1995–2004, taking into account age, sex, tumor site, tumor size, treatment used, and the years of follow-up.

Materials and methods

Data collection

A total of $N = 155$ consecutive patients diagnosed with choroid/ciliary body melanoma at the Ophthalmology Department in the Bellvitge University Hospital in Barcelona were recruited during the period from 1995 to 2004. Ten patients were excluded from this study because of incomplete data.

For each patient, a complete ophthalmologic review of both eyes was carried out, including visual acuity, slit lamp, and fundus examination. Tumor size was assessed by ocular ultrasonography and tumor classification was carried out on the basis of the Collaborative Ocular Melanoma Study (COMS) criteria: tumors were classified as small-sized melanoma if the tumor was between 1 and 3 mm in apical thickness and at least 5 mm in diameter, medium-sized melanoma when tumors were between 3.1 and 10 mm in thickness and no more than 16 mm in its longest base diameter, and large-sized melanoma when tumors were more than 10.1 mm in thickness and/or greater than 16 mm in its longest base diameter [17–19].

Patients were extensively informed about the various treatment options, such as brachytherapy (iodine-125 was the only plaque available), enucleation, and others (transpupillary thermotherapy, local resection, proton-beam radiation, and exenteration). Factors influencing the treatment chosen were the site and size of the tumor and the patient's preference for keeping the eye.

During the first 5 years after treatment, patients were checked twice a year, and during the following 5 years, they were checked only once. After 10 years of follow-up, check-up was considered voluntary. Clinical examination included blood test, chest radiograph, and liver ultrasonography to rule out distant metastases. Follow-up data on patients who did not keep their appointments were obtained by contacting the patient or their relatives by phone. Data on the sex, dates of diagnosis and treatment, treatment type, tumor location (ciliary body, choroid), longest tumor diameter, tumor thickness, extrascleral growth, tumor size (small, medium, or large melanoma), follow-up time, and eventual cause of death were collected by physicians in a structured questionnaire.

Passive follow-up was also carried out for all patients by linking their computer records with the Catalan Mortality Registry up to December 2008. However, because of the Spanish law on data protection, cause-specific mortality could not be ascertained. Permission to use the Catalan Mortality Registry was obtained by means of an agreement between our centers and the Catalan Government.

Statistical methods

Survival was analyzed using the life-table method described by Hakulinen and Abeywickrama [13,15], in which RS is used as an estimate of cause-specific mortality. RS is defined as the ratio of the observed survival (all cause survival) of a cohort of cancer patients to the expected survival of a comparable set of cancer-free individuals [15]. As a cohort of cancer-free individuals is difficult to obtain, expected life tables representing the survival of the general population were used instead. The underlying assumption is that due to cancer deaths are a negligible proportion of all deaths.

Observed survival is the probability of surviving from all causes of death for a group of cancer patients under study, and it can be estimated using the life-table method. Observed survival only takes into account whether the patient has died; it does not account for the specific cause of death. Expected survival is the survival probability of a population similar to the patient group but free of the specific disease under study [15], and it can be estimated using the national life table for individuals from the general population matched by age, sex, and calendar year of the cancer patients under the study. In our study, RS can be interpreted as an estimate of the probability of surviving the extramortality hazard related to UM. Thus, one would expect the RS to be a nonincreasing function of time. However, the RS values might increase [14] when the presence of the disease reduces the risk of death from other causes because of the close medical attention received by those patients [15]. RS yields excess hazard ratios (EHR), as opposed to the standard hazard ratios obtained in Cox and other survival models. The influence of prognostic factors on RS can be assessed by modeling and expressed as EHR; their interpretation is straightforward. For example, an EHR of 2 for men would suggest that the excess mortality rate in men (i.e. deaths associated with UM) is twice as high as in women [20].

The 5-year RS and its 95% confidence interval (95% CI) were calculated stratifying for the following prognostic factors: sex, age at diagnosis, tumor size, tumor site, follow-up period, and treatment type. EHR for these prognostic factors were modeled using a generalized linear model with the Poisson error structure [21].

Results

There were 88 women and 67 men in the study cohort. The distribution of the patients according to follow-up, age, sex, tumor size, tumor origin, treatment, and overall deaths is shown in Table 1.

The overall 5-year observed survival and RS of our patients was 84.5% (95% CI: 79–90.4%) and 90% (95% CI: 84.2–96.3%), respectively. Figure 1 shows the RS and observed survival of UM patients.

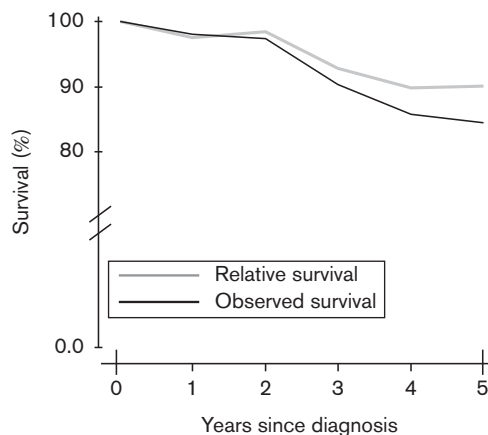
Table 2 and Fig. 2 show the 5-year survival, stratified by the variables included in the study. RS by sex shows that

Table 1 Distribution of patients with uveal melanoma according to follow-up, sex, age, tumor size, treatment, tumor origin, and deaths

Variables	N (%)		
	Both sexes	Men	Women
Years of follow-up ^a			
0–1	155 (100.00)	67 (100.00)	88 (100.00)
1–2	152 (98.06)	67 (100.00)	85 (96.59)
2–3	151 (97.41)	67 (100.00)	84 (95.45)
3–4	140 (90.32)	64 (95.50)	76 (86.36)
4–5	133 (85.80)	63 (94.02)	70 (79.54)
Sex			
Men	67 (43.30)	67 (100.00)	–
Women	88 (56.70)	–	88 (100.00)
Age			
≤ 60	75 (48.40)	36 (53.70)	39 (44.30)
> 60	80 (51.60)	31 (46.30)	48 (55.70)
Tumor size			
Small	13 (8.40)	2 (3.00)	11 (12.50)
Medium	90 (58.06)	39 (58.20)	51 (58.00)
Large	52 (33.54)	26 (38.80)	26 (29.50)
Treatment			
Brachytherapy	94 (60.65)	37 (55.20)	57 (64.80)
Small	5 (5.32)	1 (2.70)	4 (7.00)
Medium	71 (75.53)	29 (78.40)	42 (73.70)
Large	18 (19.15)	7 (18.90)	11 (19.30)
Enucleation	40 (25.81)	19 (28.40)	21 (23.90)
Small	1 (2.50)	0 (0.00)	1 (4.80)
Medium	10 (25.00)	4 (21.10)	6 (28.60)
Large	29 (72.50)	15 (78.90)	14 (66.70)
Others ^b	21 (13.55)	11 (16.40)	10 (11.40)
Small	7 (33.33)	1 (9.10)	6 (60.00)
Medium	9 (42.86)	6 (54.50)	3 (30.00)
Large	5 (23.81)	4 (36.40)	1 (10.00)
Tumor origin			
Choroid	131 (84.50)	58 (86.60)	73 (83.00)
Ciliary body	24 (15.50)	9 (13.40)	15 (17.00)
Deaths	24 (15.84)	5 (7.46)	19 (21.59)

^aNumber of patients at risk at the beginning of the follow-up interval.

^bTranspupillary thermotherapy (*n*=3), local resection (*n*=13), proton-beam (*n*=2), missing (*n*=3).

Fig. 1

Relative and observed survival of patients with uveal melanoma diagnosed during the period 1995–2004 and followed up until December 2008.

women had significantly lower RS than men during the study period (5-year RS women: 84.6%; RS men: 100%). The RS rates for patients older than 60 years were lower

Table 2 Observed and relative survival of patients with uveal melanoma according to sex, age, tumor size, treatment, and tumor origin

Variables	5-year observed and relative survival				
	Risk (N)	OS	95% CI	RS	95% CI
Global	133	84.5	79.0–90.4	90.0	84.2–96.3
Sex					
Men	63	92.5	85.7–99.3	100	94.2–109.1
Women	70	78.4	67.4–89.4	84.6	72.7–96.4
Age					
≤ 60	71	93.3	87.3–99.4	94.8	88.6–100.0
> 60	63	76.2	64.0–88.5	88.8	74.5–103.0
Tumor size					
Small	13	92.3	76.6–108.0	98.3	81.5–115.0
Medium	83	90.0	83.1–96.9	95.1	87.8–102.4
Large	40	73.1	56.6–89.6	80.8	62.6–99.0
Treatment					
Brachytherapy	85	88.3	80.9–95.7	93.6	85.8–101.4
Small	5	100.0*	–	100.0*	–
Medium	64	88.7	80.4–97.0	93.7	85.0–102.5
Large	16	83.3	62.7–104.0	88.5	66.6–110.5
Enucleation	34	72.5	53.4–91.6	80.5	59.3–101.7
Small	1	100.0*	–	100.0*	–
Medium	10	90.0	69.3–110.7	95.9	73.9–117.9
Large	21	65.5	39.1–91.9	73.6	44.0–103.3
Other	21	90.5	76.6–104.4	94.7	80.2–109.2
Small	7	85.7*	–	91.0*	–
Medium	9	100.0*	–	100*	–
Large	4	80.0*	–	83.8*	–
Tumor origin					
Choroid	114	85.5	78.4–92.5	92.4	84.8–100.0
Ciliary body	23	79.2	58.6–99.7	84.8	62.8–106.8

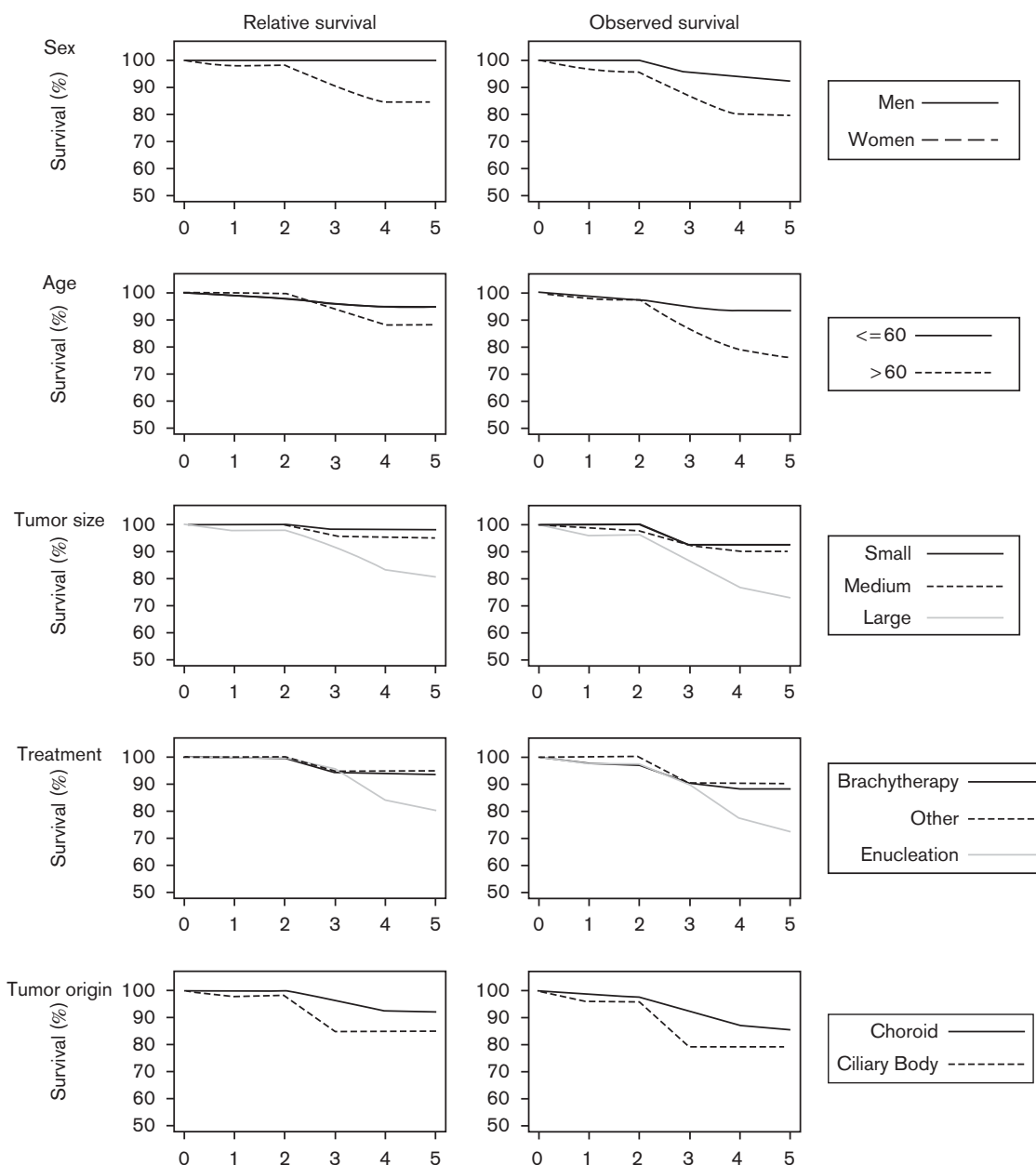
95% CI, 95% confidence interval; OS, overall survival; risk (N), number of uveal melanoma patients at risk at the beginning of the last annual interval; RS, relative survival.

*Confidence interval not calculated because less than 10 individuals were at risk at the beginning of the last annual interval.

than those younger than 60 years (5-year RS >60: 88.8%; ≤ 60: 94.8%), but the difference was not significant. RS by tumor size showed that patients diagnosed with large-sized melanomas had significantly lower survival (5-year RS: 80.8%) than those diagnosed with small-sized (5-year RS: 98.3%) and medium-sized melanomas (5-year RS: 95.1%). Overall, patients treated with enucleation showed significantly lower RS rates (5-year RS: 80.5%) than those treated with brachytherapy (5-year RS: 93.6%) or other treatments (5-year RS: 94.7%). It should be noted that no significant differences were found among medium-sized melanoma patients treated with brachytherapy (5-year RS: 93.7%) compared with the enucleated ones (95.9%) or with those whom received other therapies (100%). In contrast, there were no statistically significant differences between intraocular tumor origin (5-year RS: 92.4% choroid; 84.8% ciliary body).

The RS regression model (Table 3) describes the simultaneous effects of age, sex, follow-up years, melanoma size, treatment, and tumor origin. The first column shows the effect of each variable adjusted by follow-up, whereas the second shows the effect of each variable adjusted by follow-up, sex, and age, and the third column shows the effect of each variable adjusted by the

Fig. 2



Relative and observed survival of uveal melanoma by sex, age, tumor size, treatment, and tumor origin.

remaining ones. Tumor origin was not found to be statistically significant in any of the regression models. In contrast, a significant excess mortality among UM patients was related to sex (EHR women: 3.65), older age (EHR > 60: 2.25), large-sized melanoma (EHR: 2.45), enucleation (EHR: 1.67), and third and fourth years of follow-up (EHR: 5.37 and 3.01, respectively).

Discussion

This is the first Spanish study in a single referral center reporting the 5-year RS among UM patients that takes into account patients' features as well as tumor origin.

The overall 5-year RS from UM was 90%, higher than those reported in other population-based studies carried out in Sweden (70.1% at 5 years and 59% at 10 years) [22], USA (close to 80%) [23], and England and Wales (72%) [24]. The higher overall RS found in this study could be attributed in part to a possible selection bias because UM cases were previously selected to receive conservative treatment in a referral single hospital.

According to tumor size, our patients showed the following 5-year RS: 98.3% (95% CI: 81.5–115%) for small tumors, 95.1% (95% CI: 87.8–102.4%) for medium

Table 3 Multivariate 5-year relative survival by follow-up, sex, age, tumoral size according to COMS, treatment received, and tumor origin

	95% CI		
	EHR ^a	EHR ^b	EHR ^c
Follow-up			
[0;1)	Ref.	Ref.	Ref.
[1;2)	0.05 (0.02; 0.08)*	0.09 (0.05; 0.17)*	0.13 (0.07; 0.24)*
[2;3)	4.71 (2.32; 9.53)*	4.94 (2.36; 10.31)*	5.37 (2.63; 10.95)*
[3;4)	3.10 (1.48; 6.48)*	3.24 (1.50; 7.01)*	3.01 (1.42; 6.36)*
[4;5)	1.52 (0.73; 3.18)	1.47 (0.68; 3.17)	1.01 (0.49; 2.06)
Sex			
Men	Ref.	Ref.	Ref.
Women	2.92 (2.22; 3.86)*	3.17 (2.20; 4.57)*	3.65 (2.43; 5.46)*
Age			
≤ 60	Ref.	Ref.	Ref.
> 60	2.81 (1.46; 5.42)*	2.79 (1.55; 5.02)*	2.25 (1.24; 4.07)*
Tumor size			
Small-medium	Ref.	Ref.	Ref.
Large	2.98 (2.07; 4.30)*	3.23 (2.19; 4.76)*	2.45 (1.53; 3.94)*
Treatment			
Brachytherapy and other	Ref.	Ref.	Ref.
Enucleation	2.87 (1.98; 4.16)*	2.91 (1.92; 4.39)*	1.67 (1.04; 2.68)*
Tumor origin			
Choroid	Ref.	Ref.	Ref.
Ciliary body	1.07 (0.77; 1.49)	1.11 (0.74; 1.65)	1.18 (0.72; 1.93)

95% CI, 95% confidence interval; COMS, Collaborative Ocular Melanoma Study; EHR, excess hazard ratio; Ref, reference category for the corresponding variable.

^aEHR adjusted by follow-up.

^bEHR adjusted by follow-up, sex, and age.

^cEHR adjusted by follow-up, sex, age, treatment, and tumor origin.

*Statistically significant at $\alpha=0.05$.

tumors, and 80.8% (95% CI: 62.6–99%) for large tumors. When these results were compared with melanoma-specific survival published in the COMS study (99% for small, 89–91% for medium, and 71–72% for large melanomas) we observed that small-sized melanoma patients had similar mortality in both series, whereas our medium-sized and large-sized melanoma patients presented slightly better prognoses [16,18,19].

In our study, one of the lowest 5-year RS rates was found for enucleated patients (RS: 80.5%). RS was 93.6 and 94.7% for those who received brachytherapy or other treatments, respectively. The most feasible explanation is that 72.5% of the enucleated patients had a large tumor size, whereas 80% of the patients who were not enucleated had small-sized or medium-sized UM. It is well known that tumoral size is a risk factor for death [25,26]. In contrast, medium-sized UM patients did not show statistically significant differences when they were compared by treatment. It has been largely reported that tumoral medium-sized UM treated with enucleation or brachytherapy have similar survival rates [4,9,18].

We observed a peak in excess mortality during the third and fourth year after diagnosis, even adjusting for age, sex, tumor size, treatment, and tumor origin, suggesting that it is necessary to reinforce the follow-up of patients during the first 5 years, particularly after the third year. This excess mortality was previously related to the enucleation procedure; at present, it has been explained on the basis of early UM metastasis [27]. In a population-based survey in Sweden, 2997 UM cases showed

statistically significant excess mortality at 3 years of follow-up; this increase in mortality remained up to 5.5 years [22]. In this study, patients diagnosed more recently have a better 5-year RS, but we should not draw conclusions on the impact of eye-sparing treatments on survival, as the patients' survival could not be adjusted for different treatments or for UM stage. In the same manner, the COMS study confirms the presence of a higher initial mortality. Follow-up after treatment for choroidal melanoma showed that mortality was independent of the treatment method (enucleation or plaque therapy). The annual mortality rate in both groups was similar, with a mortality rate of 5–6% in the second and third years of follow-up, followed by a slow decline of about 1% after 9 years of follow-up [4]. This excess mortality in the first year after UM diagnosis needs further investigation to identify patients with an elevated risk for UM metastasis and to offer preventive adjuvant therapy.

We show in our study that UM mortality is similar to the Catalan population's expected survival in small-sized and medium-sized melanoma (on the basis of the survival Catalan life expectancy tables) because the 5-year RS among these patients was 98 and 95%, respectively. Despite these results, the potential metastasis of UM reported several decades after diagnosis has been taken into account. The probability of being alive without metastases after 25 years was found to be 40%, assuming that the patient did not die from any other cause. For these reasons, a long follow-up period is important, as melanoma could reappear 40 years after diagnosis [5,7,8].

Patients in the large tumor size category showed the lowest RS rates and the highest risk of death (EHR: 2.45) when compared with patients with small and medium tumor sizes, after adjusting for age, sex, follow-up, treatment, and tumor origin. It is well known that the largest tumors are associated with an increased risk of death after enucleation [2], brachytherapy [28,29], helium ion [30] and proton-beam [31], and are a prognostic factor for recurrence [5,27,32].

In our study, patients older than 60 years of age showed an increased risk of death during the study period (EHR: 2.25, $P < 0.05$) compared with younger patients, taking into account the effects of the other prognostic factors. These results are in agreement with other studies [2,4,22]. However, it is not clear as to what extent this correlation is the result of bias arising from competing risks [6].

In our patients, women had a 3.65-fold higher EHR of death than men taking into account the effects of the other prognostic factors, showing that survival was higher in men than in women. This finding is not in agreement with other studies, which found better survival for women than for men [4,33]. As shown in several series, UM is slightly more common in men, but it is unclear whether this is a primary effect of sex or a secondary effect related to occupational or recreational exposure [34–37], and we could not identify referral patterns in our hospital that explained this specific behavior.

In our study, we found that tumor origin was not a prognostic factor in UM patients after adjusting for age, sex, follow-up, and tumor size. These results are in disagreement with other studies that show ciliary body involvement to be a risk factor for death [2,5].

Some limitations must be taken into account to accurately interpret the results of this report: (i) selection bias may have occurred because patients were recruited from a single referral hospital of Catalonia and (ii) the small number of cases included in the study could reduce the study's statistical power, resulting in conservative conclusions. It is noteworthy that this is not a study-based population as most published series, but a study in a single referral center and that clinical factors other than age and sex were evaluated unlike the previous series [22,23,29,33,38].

In conclusion, and despite these limitations, this is the first Spanish study in a single center reporting RS among UM patients and revealed low 5-year mortality rates in our UM patients, supporting our current management practice with conservative treatments. Periodical estimates of RS in the UM patients allow us to compare mortality between different centers, without the need to know the exact cause of death, and therefore without the risk of misclassification of patients who died.

In our study, the probability of death is higher among women, older patients, patients with large-sized melanoma, and enucleated UM patients, whereas tumor origin does

not seem to play a role in the determination of excess mortality. It is noteworthy that the significant excess of mortality during the third and fourth years of follow-up reinforces the need for a strict follow-up of UM patients during this period of time.

Acknowledgements

Conflicts of interest

There are no conflicts of interest.

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